

TyreVibes - App Summary (Repo-Based)

What it is

TyreVibes is an iOS SwiftUI app for tyre and vehicle maintenance management with tyre wear/tread analysis, history tracking, and reminders. Repo evidence also shows a hybrid backend (Supabase plus custom Node API).

Who it's for

Primary persona: Individual car owners seeking automated tyre and maintenance tracking (conductor/product.md).

What it does

- Garage management for vehicles and tyre sets (Garage and Tyres feature screens).
- Tyre analysis workflows, including image-based and LiDAR-based tread measurement services.
- Authentication with email/password, OTP, Apple Sign-In, and Google Sign-In via Supabase Auth.
- Maintenance planning/history with scheduling and reminder-related services.
- Notification system with local scheduling, push handling, and deep-link routing.
- Cloud persistence through Supabase tables plus REST endpoints under /api/v1/*.
- Report/export capabilities (PDF/CSV builders and report screens in-app).

How it works (compact architecture)

- Client: SwiftUI app entry in TyreVibesApp.swift; feature screens under TyreVibes/Features/*.
- Service layer: AuthService, Tyre/Vehicle services, LiDAR/CV analyzers, notification and scheduler managers.
- Data/auth: SupabaseManager initializes the Supabase client from Api.plist (URL/key).
- API path: RemotePersistenceService posts tyre and analysis payloads via NetworkManager to /v1 endpoints.
- Backend: server.js (Express + JWT + MySQL + Sharp) and Supabase Edge Functions for periodic jobs.
- Background flow: BackgroundTaskManager calls edge functions, syncs notifications, and reschedules refresh.

How to run (minimal getting started)

- Open /Users/matteolamannna/Desktop/TyreVibes/TyreVibes/TyreVibes.xcodeproj in Xcode.
- Verify TyreVibes/Api.plist has required keys for Supabase and Google auth integration.
- Set signing/capabilities for your Apple team and run the TyreVibes target on simulator/device.
- Optional backend jobs: follow supabase/README.md or DEPLOYMENT_GUIDE.md for edge-function deployment.
- Local backend start command for server.js: Not found in repo (no package.json/scripts discovered).